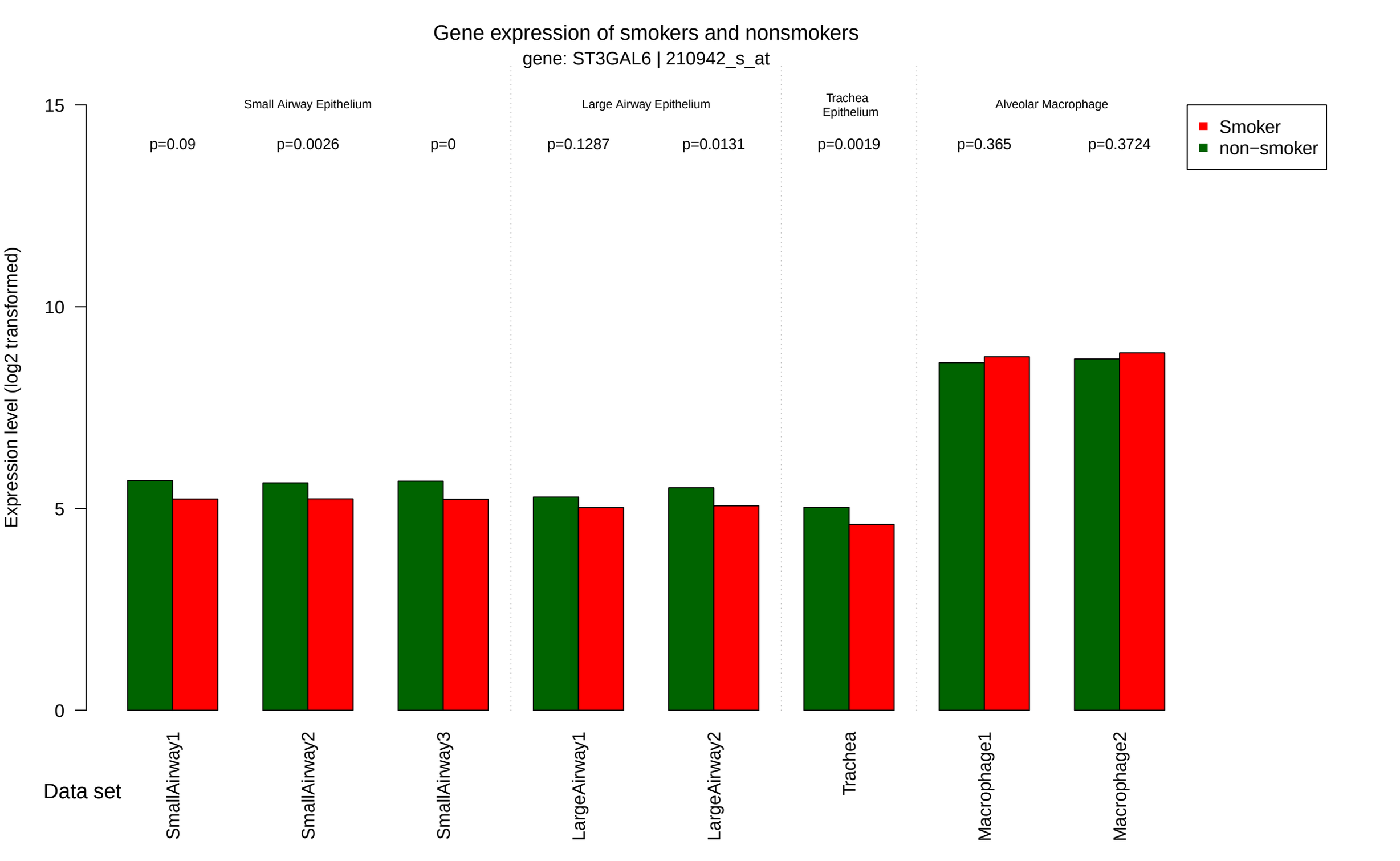
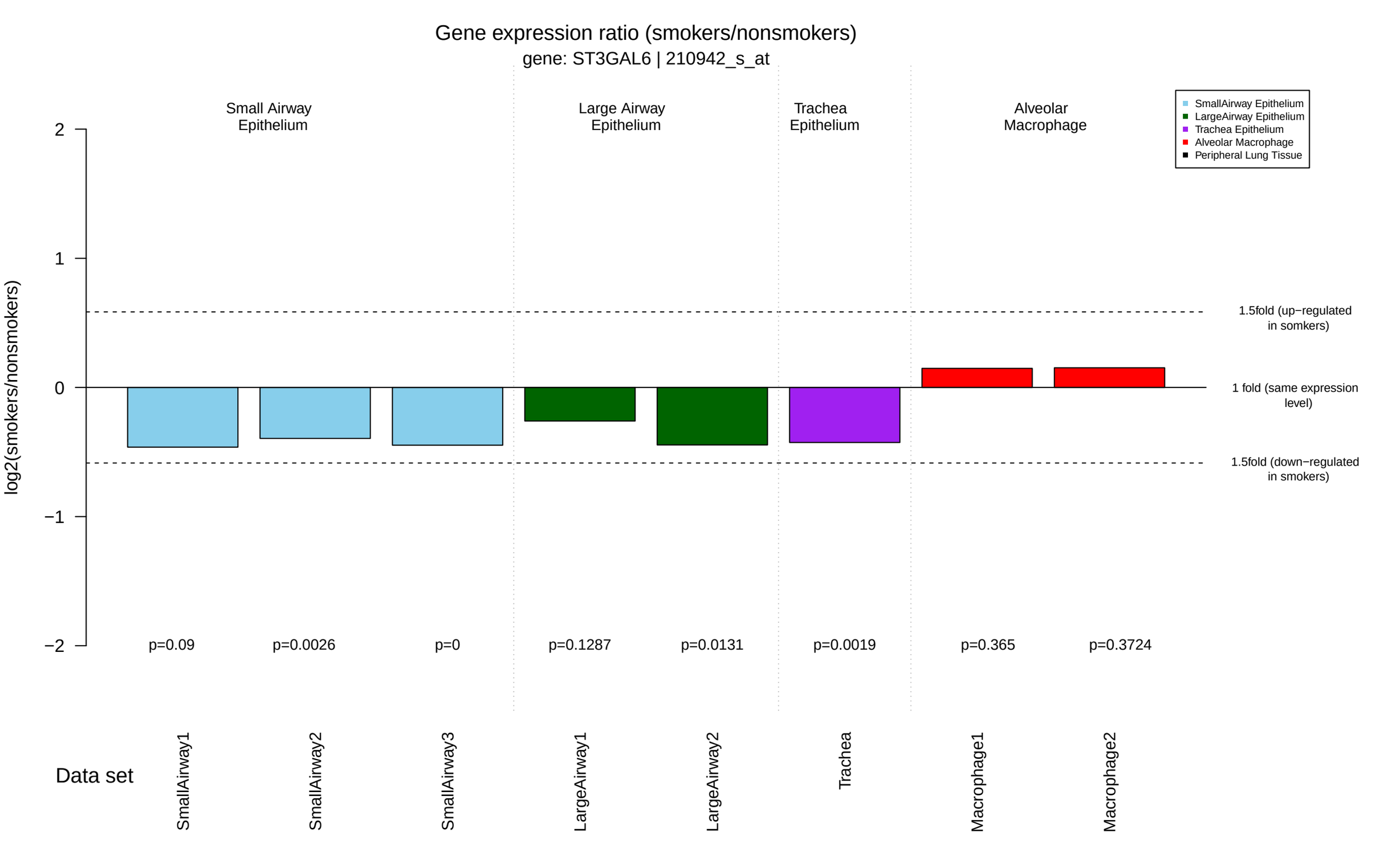
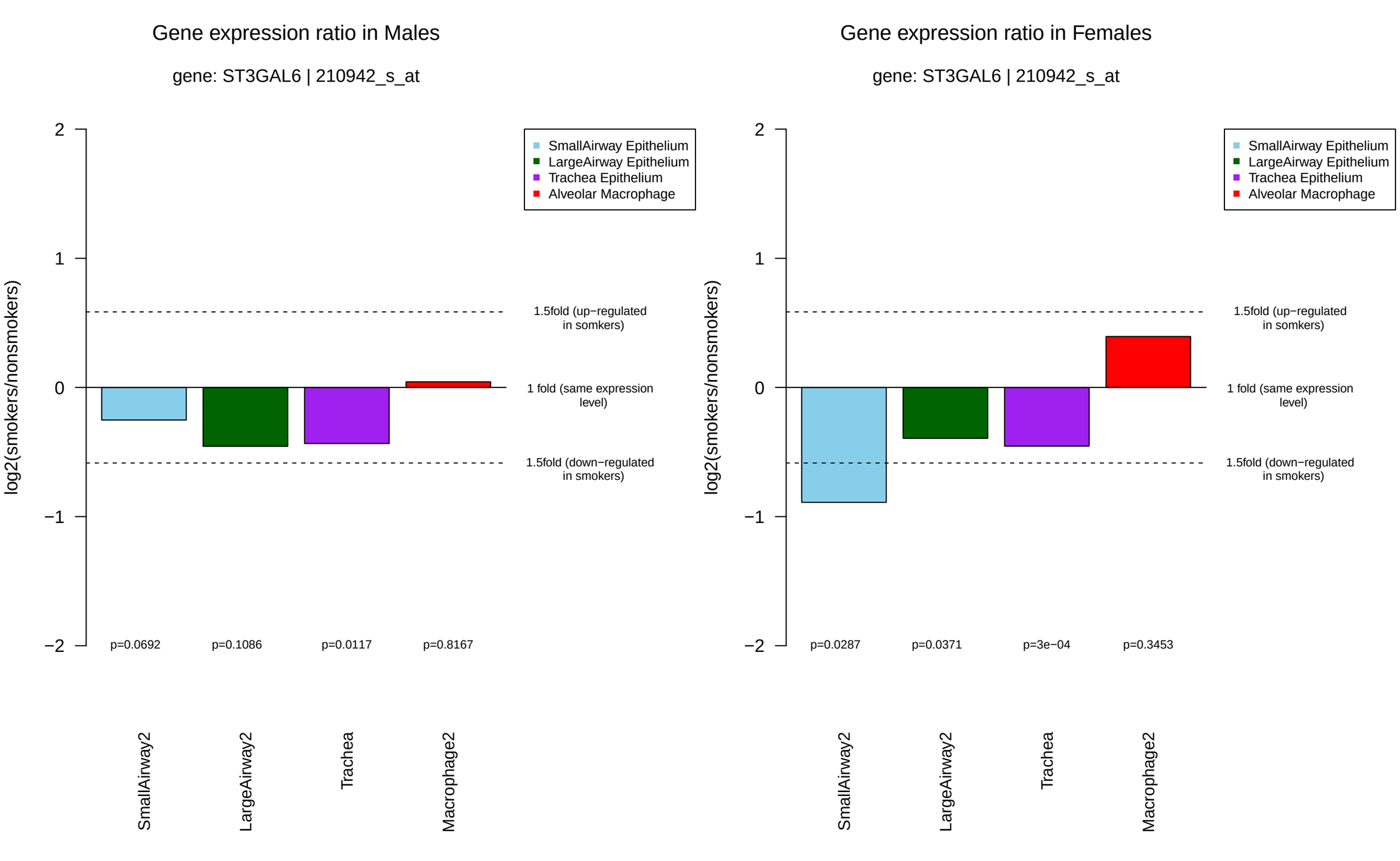
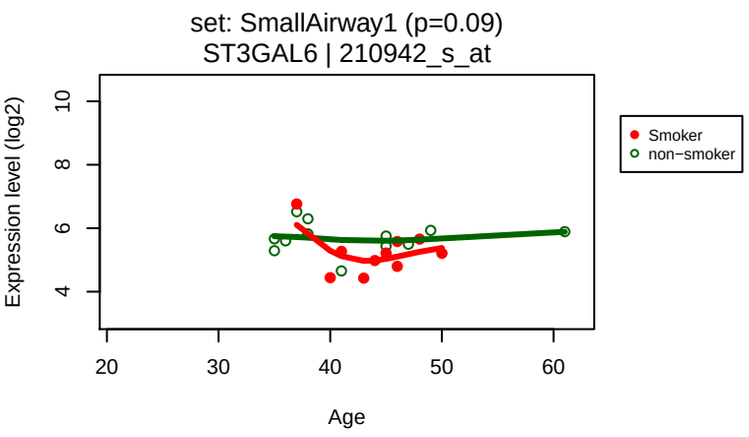
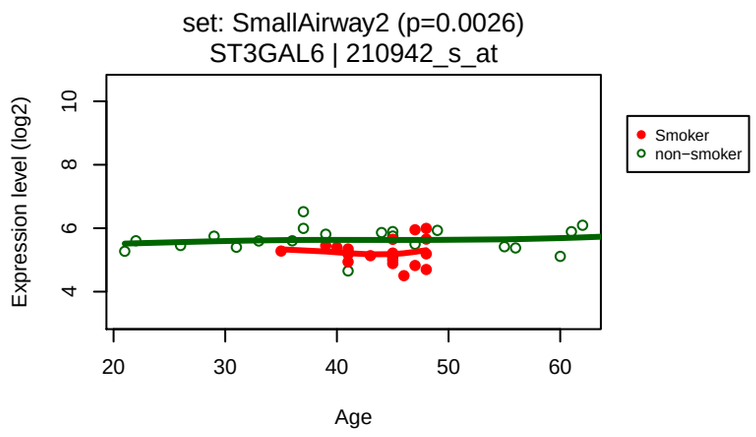
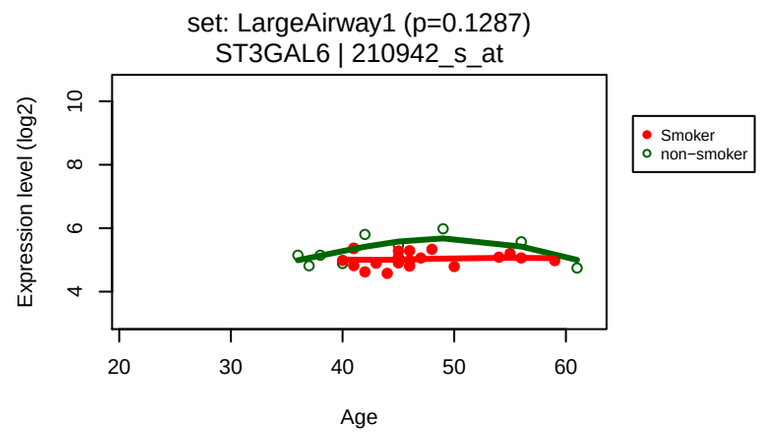

Gene expression of smokers and nonsmokers
gene: ST3GAL6 | 210942_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.09
p=0.0026
p=0
p=0.1287
p=0.0131
p=0.0019
p=0.365
p=0.3724
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

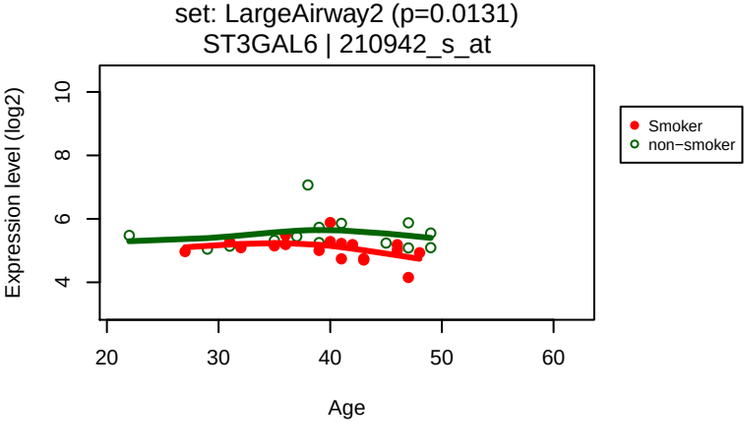

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL6 | 210942_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.09
p=0.0026
p=0
p=0.1287
p=0.0131
p=0.0019
p=0.365
p=0.3724
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

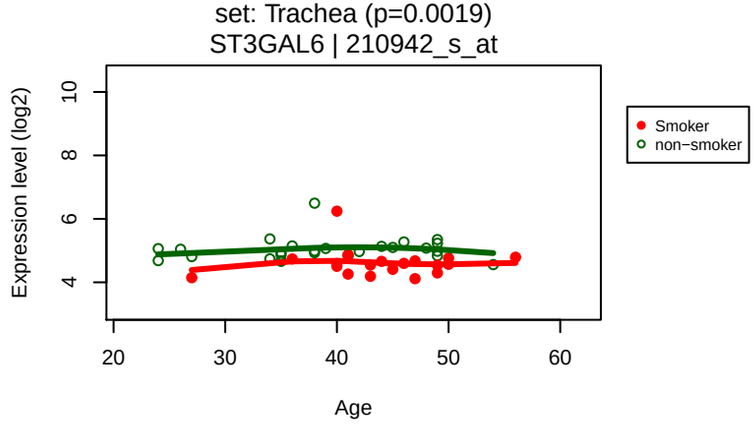

Gene expression ratio in Males
gene: ST3GAL6 | 210942_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0692
p=0.1086
p=0.0117
p=0.8167
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL6 | 210942_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0287
p=0.0371
p=3e−04
p=0.3453
SmallAirway2
LargeAirway2
Trachea
Macrophage2

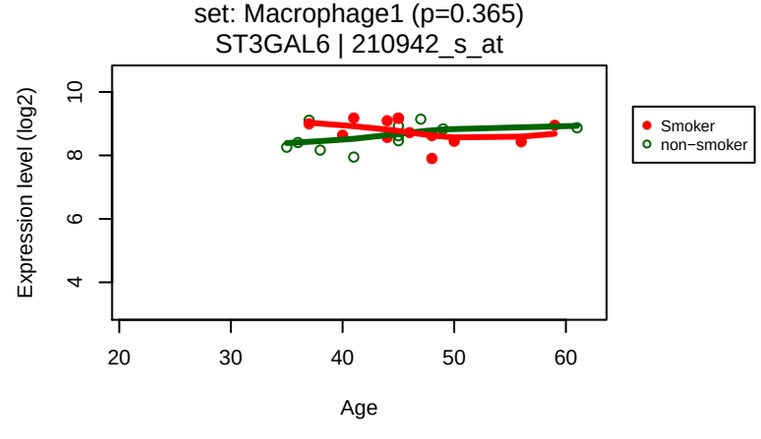

set: SmallAirway1 (p=0.09)
ST3GAL6 | 210942_s_at
Expression level (log2)
Age
Smoker
non−smoker

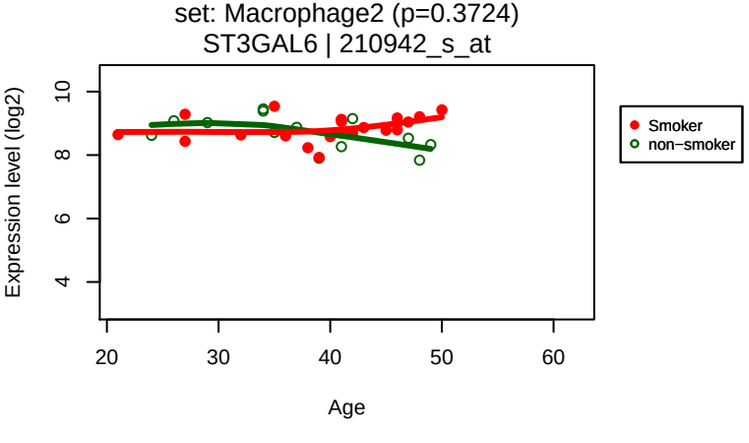

set: SmallAirway2 (p=0.0026)
ST3GAL6 | 210942_s_at
Expression level (log2)
Age
Smoker
non−smoker

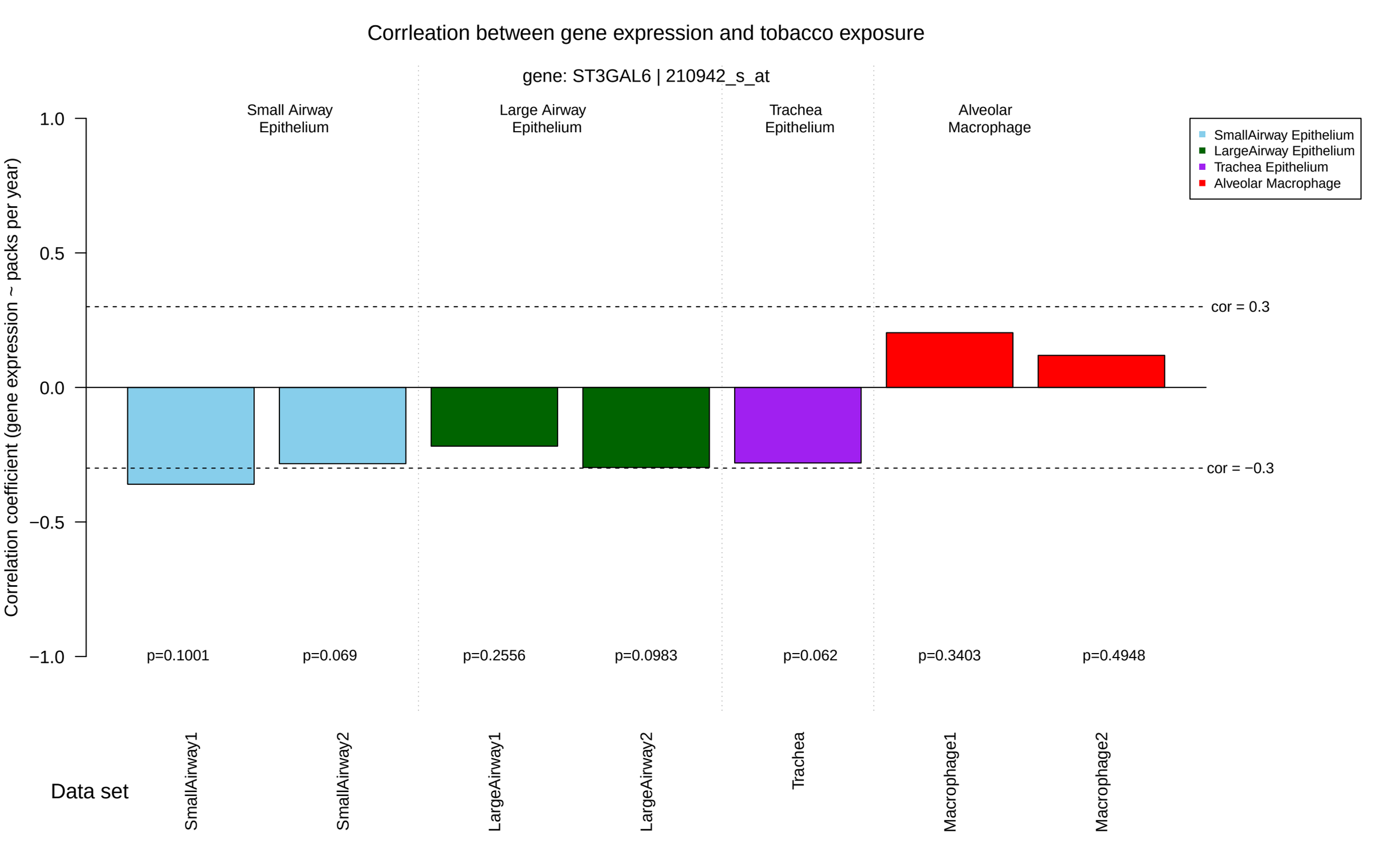

set: LargeAirway1 (p=0.1287)
ST3GAL6 | 210942_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0131)
ST3GAL6 | 210942_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0019)
ST3GAL6 | 210942_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.365)
ST3GAL6 | 210942_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3724)
ST3GAL6 | 210942_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL6 | 210942_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1001
p=0.069
p=0.2556
p=0.0983
p=0.062
p=0.3403
p=0.4948
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2